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tutorialverse: Interoperability and quality infrastructure for R tutorials

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Executive summary

Instructors and learners choosing an R tutorial often have to inspect its repository to find prerequisites and evidence that the exercises still work. Descriptions differ across formats and may leave maintenance status unclear.

tutorialverse will help instructors and learners find R tutorials suited to their needs and see the reported prerequisites, check results, and review status. Authors will describe tutorials with shared metadata. An R package will use adapters to run supported checks through existing engines and produce common reports. A searchable registry will link shared metadata and validation records to original lessons.

An [initial prototype](#) already demonstrates the proposed adapter model across all three engines. Grant work will extend this foundation to curated registry records, CI-based validation, author-effort measurement, and community-reviewed metadata.

Over 12 months, we will harden and extend support for `learnr`, `learnr2`, and Quarto Live, index approximately 15 existing tutorials, and use four small reference fixtures to test the adapters across teaching contexts. Carpentries-style materials will also be indexed, with stated limits on automated checks. Existing tools will provide browser execution and static hosting. We request USD 10,000 for this work.

Signatories and community support

Shaurita D. Hutchins will lead the project in collaboration with Samuel Bharti. Both are PhD candidates at the University of Alabama at Birmingham.

Hutchins will lead educational design and community development. She will develop the metadata profile, coordinate reference fixtures, and establish accessibility and contribution guidance. She brings experience in Carpentries instruction, reproducible R/Bioconductor workflows, and [browser-based educational modules](#).

Bharti will lead software architecture and engine integration. He will harden and extend adapters that use existing engine checks and produce validation reports in a common format. A former Posit intern, he develops Shiny applications, R packages, and automated documentation workflows. His teaching work includes [r-for-beginners](#), an adaptation of Software Carpentry lessons with accompanying slides.

We will jointly design the package API and registry, and share responsibility for governance and the reference fixtures.

We requested public feedback through [r/Stats](#) and the [learnr webR discussion](#). Community contributor [@coatless](#) pointed us to `learnr2` and Quarto Live. That feedback helped us focus on shared metadata and checks, though the proposed profile still needs review.

We are seeking additional review from engine maintainers, Carpentries and Bioconductor educators, and university and community instructors.

Problem

Learners need tutorials that match their goals and current knowledge; instructors need lessons they can assess and reuse in their teaching. Both need clear information about content, prerequisites, difficulty, and duration. They also need evidence that the exercises still work, since a tutorial may remain online after they break or its author stops maintaining it.

The tools used to build a lesson also affect how it runs. `learnr` runs tutorials as Shiny applications (Aden-Buie et al. 2025). `learnr2` builds on Quarto Live, which uses `webR` to execute R in the browser (*learnr2*, n.d.; Stagg 2026). `tutorialverse` will give instructors and learners a common way to compare tutorial descriptions and check results across these engines.

Evidence of need and prior work

Choosing tools for an R tutorial involves practical tradeoffs. Handel’s (2025) comparison of six tools for Quarto exercises shows the hosting, maintenance, and exercise-support decisions tutorial authors face. These differences also affect reuse: instructors need to know a tutorial’s requirements and maintenance status before adopting it. `tutorialverse` will make that information easier to find and compare.

For instructors and learners seeking existing tutorials, discovery presents a related challenge. `rstats-ed` addressed it through a community-curated catalog, while acknowledging incomplete coverage. `Glitr.org` helps users find and compare teaching repositories, but excludes educational-quality evaluation (Geest et al. 2024). Consistent descriptions support discovery: `TeSS` aggregates training metadata, and `Bioschemas` provides shared specifications for resource descriptions (Garcia et al. 2020; Castro et al. 2023).

Assessing a resource for reuse requires evidence beyond its description. The Galaxy Training Network combines automated checks with community review and maintenance (Hiltemann et al. 2023). Formal scholarly review serves a different purpose: the *Journal of Open Source Education* (JOSE) peer-reviews reusable educational materials and software (Journal of Open Source Education, n.d.). `tutorialverse` will distinguish these external reviews from technical check results.

`tutorialverse` will bring these practices together across R tutorial engines, linking shared descriptions to validation results, review history, and maintenance information. Existing engines will remain responsible for execution. New engines, browser runtimes, and learning-management systems are outside scope; static hosting will be described and checked as a deployment mode.

Proposal

The five milestones follow a tutorial from description through checking, discovery, and review.

Milestone 1: A minimal metadata profile

The pilot profile currently requires eight fields: identifier, title, source URL, lesson URL, engine, maintainer, license, and topics. Learning objectives are currently optional; community review will determine whether topics or learning objectives belong in the minimal required set. We will map fields to Schema.org and the Bioschemas TrainingMaterial profile where applicable. Level, prerequisites, packages, deployment mode, duration, accessibility information, and software versions will remain optional initially. Optional review references and DOIs will identify external reviews and publications. The checking tools will generate validation dates and results.

To reduce manual work, we will evaluate drafting metadata from existing tutorial sources. Suggested topics, prerequisites, learning objectives, and difficulty will require human confirmation before publication. Records will distinguish author-supplied values, mechanically extracted information, and inferred suggestions, with review status recorded separately. We will prefer deterministic extraction and heuristics where possible; AI assistance will be optional. The assisted-metadata pilot will evaluate author effort using the measures in Milestone 4.

Authors will supply missing fields in a small accompanying file without rewriting lessons. Unknown values will remain explicit, and nominations with unknown maintainers or licenses will remain pending review.

We will align the prototype with this draft and invite engine maintainers and educators to review it. Their feedback and the pilot’s measures of author effort will guide revisions before we publish version 1.0 with example records and adoption guidance.

Milestone 2: Harden and extend engine adapters

The prototype implements `learnr_adapter()`, `learnr2_adapter()`, and `quarto_live_adapter()`, producing a common report through `check_tutorial()`. We will harden and extend these adapters, refine their interfaces with maintainers, and document supported versions and features. Because `learnr2` and Quarto Live share formats, adapters cannot always distinguish them automatically.

Adapters may invoke an engine's existing build or checking tools; the engine remains responsible for execution. Conversion between engines is outside the adapters' scope. A capability table will show each adapter's support for metadata import, rendering checks, code execution, and accessibility checks. Engines can support different checks while reporting results in the same format. Carpentries-style materials can be indexed after metadata and link checks. Further checks will depend on the lesson's build system, with those limits stated in the registry.

Milestone 3: Validation across engines

`check_tutorial()` already combines metadata, URL, and optional rendering checks using four statuses: `pass`, `fail`, `not_run`, and `unsupported`. Grant work will add deeper engine-specific checks of required files, declared dependencies, package versions, interactive execution, and exercise identifiers where supported. Basic automated accessibility checks will accompany manual keyboard checks. Reports will distinguish observed results from declared accessibility information.

We will automate reproducible checks and investigate browser tests of representative exercises and assessments. Successful rendering will remain distinct from evidence that interactive behavior works. Reports will identify failing fields or exercises and record the test environment, source version, and date. Automated checks alone will not establish teaching quality or full accessibility.

We will publish a reusable GitHub Actions workflow and integration guidance so authors can run checks in their own repositories without joining the registry. CI will test the package on Linux, macOS, and Windows, using examples with known faults to verify that the checks detect problems. Documented real renders currently cover macOS only.

Milestone 4: A searchable tutorial registry

The registry will link tutorial descriptions to the original repositories and published lessons. Authors will retain responsibility for content and hosting. Automated builds will use YAML entries to create the searchable website and publish `index.json`. A proposed `tutorial_search()` function will return results as an R data frame.

Readers will filter by topic, package, level, learning objective, engine, and deployment mode when those fields are available. Each entry will show duration and maintenance status when known, and distinguish metadata validity from execution checks and human review. Scheduled checks will flag broken links and outdated validation records.

The indexing pilot will cover 5–10 existing tutorials across the supported formats, expanding to approximately 15 indexed tutorials by the end of the grant. In the assisted-metadata pilot, we will count fields automatically extracted, suggested, and manually entered; record whether suggestions are accepted, edited, or rejected; and measure minutes to produce a valid record. We will compare assisted and manual preparation. Pilot completion will require valid records and working search results; claims about execution will depend on separate adapter tests.

Milestone 5: Review, governance, and reference fixtures

Authors will submit entries through structured pull requests; others can nominate tutorials. Drawing on [rOpenSci's review process](#) and Carpentries lesson-review practices ([rOpenSci, n.d.](#); [The Carpentries 2022](#)), automated checks will precede human verification of metadata, provenance, licensing, maintenance information, and declared accessibility. We will record external pedagogical or scholarly reviews rather than reproduce them. Registry entries will distinguish review history from current validation and maintenance status. Governance guidance will assign maintenance responsibilities and explain how to contact maintainers, correct or remove entries, and archive stale resources.

We will use four small reference fixtures, consisting of representative tutorial exercises, to test the adapters: `renv`, package development with `usethis` and `devtools`, an introductory DESeq2 workflow, and a Carpentries R companion. We will adapt existing material under compatible licenses. Bharti's *r-for-beginners* lessons and Bioconductor training resources offer starting points.

These fixtures will test metadata mapping, validation reports, and each engine’s checks; the indexing pilot will test how existing tutorials are described and found. The four fixtures will collectively cover all three adapters, without requiring every tutorial to run under every engine. Tests will include browser interaction where supported and local or server execution where needed. Three or four small integration templates will show authors how to connect existing lessons through their chosen engine.

Project plan

Preliminary work

We have implemented an initial tutorialverse prototype supporting learnr, learnr2, and Quarto Live. The package validates common tutorial metadata, checks source and published URLs, and uses engine adapters to render trusted local tutorials while producing a common validation report. The [adapter review](#) documents successful renders of one public tutorial per engine on macOS. Interactive exercise execution and accessibility remain explicitly unsupported rather than being inferred from successful rendering. Cross-platform CI, curated registry records, and measurements of author effort remain proposed work.

Schedule and working practices

Period	Work and completion criteria
Months 1–2	Review the eight-field profile, map metadata from existing tutorials into pilot records, establish repositories and governance, and begin the reference fixtures.
Months 3–5	Harden the three adapters and extend <code>check_tutorial()</code> . Publish a capability table and common report format. Test valid examples and known failures in Linux, macOS, and Windows CI.
Months 6–8	Build registry ingestion and search. Publish <code>index.json</code> and the R search interface. Run the pilot records through the package and document author effort.
Months 9–10	Pilot nominations and human review with external contributors. Test stale-entry handling and gather usability feedback.
Months 11–12	Expand the registry to approximately 15 tutorials. Complete reference-fixture and integration-template checks. Address high-priority feedback and release version 1.0 of the package, specification, and registry with maintenance guidance.

Development will occur openly on GitHub with pull-request review, automated testing, documented releases, and community updates. Code will use MIT; new educational content will use CC BY 4.0 where source licenses permit.

Budget and funding

Requested budget: USD 10,000, for development, documentation, and review across the five milestones. No hardware, cloud, travel, or fee costs are included.

Milestone	USD
Minimal metadata profile	1,500
Engine adapters	2,500
Validation package	2,750
Tutorial registry	1,750
Review, governance, and reference fixtures	1,500
Total	10,000

If funding is reduced, the profile and validation tools can be used independently of the registry.

Risks and maintenance

The prototype demonstrates the adapter model; adoption remains the main uncertainty. Authors may find the profile burdensome if it duplicates metadata they already maintain. The next experiment is the 5–10-tutorial

registry pilot: we will record which fields are reused, supplied manually, or unknown, and measure author effort. These findings and early consultation will guide revisions before version 1.0.

Engine changes could break adapters or expose differences in test coverage. We will document supported versions and browser or server requirements, and use the reference fixtures to detect regressions.

We will document contribution and governance procedures so others can maintain the registry without relying on us for every change. Entries will name maintainers and retain source links, validation dates, and status. Scheduled checks and an archival policy will help contributors address stale resources.

Success

Completion requires version 1.0 of the metadata profile, R package, and searchable registry. Deliverables will include three hardened adapters tested in Linux, macOS, and Windows CI; approximately 15 indexed tutorials; and four reference fixtures tested in CI, collectively covering all three adapters. We will publish three or four integration templates and at least one reusable validation workflow.

At least three external contributors will review the profile, adapters, or tutorials. The registry pilot will test nomination, review, and stale-entry procedures. We will document check results and reviewer feedback, track external submissions, and measure author effort, including whether assisted metadata drafting reduces manual work.

Future work could support additional engines and educational-catalog integrations. tutorialverse may provide scaffolding to help authors adapt existing vignettes, articles, or lessons into supported tutorial formats, but automatic conversion of educational content is outside the initial funded scope. Multilingual metadata, richer accessibility checks, and instructor collections remain outside the funded scope.

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